THE CURVES STACK IS A STACK IN GROUPOIDS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Equation equation-curves-over-proper-spaces (canonical tag 0D50). cited at proof line 4818. The embedding of the curves stack.
- Lemma lemma-spaces-stack (canonical tag 0D1G). cited at proof line 4821. The spaces stack is a stack in groupoids.
- Lemma spaces-morphisms-lemma-dimension-fibre-after-base-change (canonical tag 04NS). cited at proof line 4832. Relative dimension <= 1 detected on fibres.

Source reference. Original equation 0D50: sources/stacks-chow/tags.txt, source lines 16470–16470. Cited at proof line 4818: the embedding.

Source reference. Original lemma 0D1G: sources/stacks-chow/tags.txt, source lines 16346–16346. Cited at proof line 4821: the spaces stack fact.

Source reference. Original lemma 04NS: sources/stacks-chow/tags.txt, source lines 6090–6090. Cited at proof line 4832: relative dimension detected on fibres.

Lemma 0.2. *The category $\mathcal{C}urves$ is a stack in groupoids over Sch_{fppf}.*

Proof. Using the embedding (0D50), the description of the image, and the corresponding fact for $\mathcal{S}paces'_{fp,flat,proper}$ (Lemma 0D1G) this reduces to the following statement: Given an object $X \to S$ of $\mathcal{S}paces'_{fp,flat,proper}$ and an fppf covering $\{S_i \to S\}_{i \in I}$ the following are equivalent:

(1) $X \to S$ has relative dimension ≤ 1, and
(2) for each i the base change $X_i \to S_i$ has relative dimension ≤ 1.

This follows from Morphisms of Spaces, Lemma 04NS. □

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